avoid accident but also to extent mutual road courtesy.




### (a) Types of Sight Distances

i. Crossing sight distance
ii. Non-passing sight distance
iii. Passing sight distance
iv. Lateral sight distance

### iv. Road Gradient

It is the rate of rise or fall of the road along its alignment. It is expressed as the ratio of difference in height of its extremes to the horizontal length between them.

$$\text{Road gradient} = \frac{\text{Difference in Height}}{\text{Horizontal Length}}$$

### (a) Factors affecting gradient

i. Topography of the country
ii. Characteristics of the traffic
iii. Rainfall in the locality
iv. Drainage
v. Safety

### (b) Types of gradient

i. Maximum gradient
ii. Minimum gradient
iii. Average gradient
iv. Ruling gradient
v. Exceptional gradient
vi. Floating gradient

### v. Road Curves

Road curves are regular bends in roads to bring a gradual change in direction.



### (a) Classification of road curves

In general, there are two types of curves, they are

i) Horizontal Curves
ii) Vertical curves

**Horizontal Curves:** It is a curve in plain provided to change the direction of alignment. Horizontal curves are of different types as follows

- Simple circular curve
- Compound curve
- Reverse curve
- Transition curve
- Spiral
- Lemniscate

**Vertical Curves:** It is a curve in elevation provided at change of gradients. Vertical curves are of two types, namely

- Summit curve or crest curve
- Valley curve or sag curve



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